

Multi-Sample Comparison Report

Report Date: July 10, 2026 at 10:30

Number of Samples: 51

Temperature: 20 °C

Porosity: Varies by sample (0.267 to 0.434)

Executive Summary

Comparison Analysis: This report compares 51 soil samples based on grain size distribution and hydraulic conductivity estimates.

Key Findings:

- Highest permeability: B102_21 (7.93e-04 m/s)
- Lowest permeability: B101_13 (4.96e-05 m/s)
- Permeability range: 16.0x difference

Methodology

Comparative Analysis Approach

This comparison report presents a side-by-side analysis of multiple soil samples to identify patterns, variations, and relationships between grain size characteristics and hydraulic conductivity estimates.

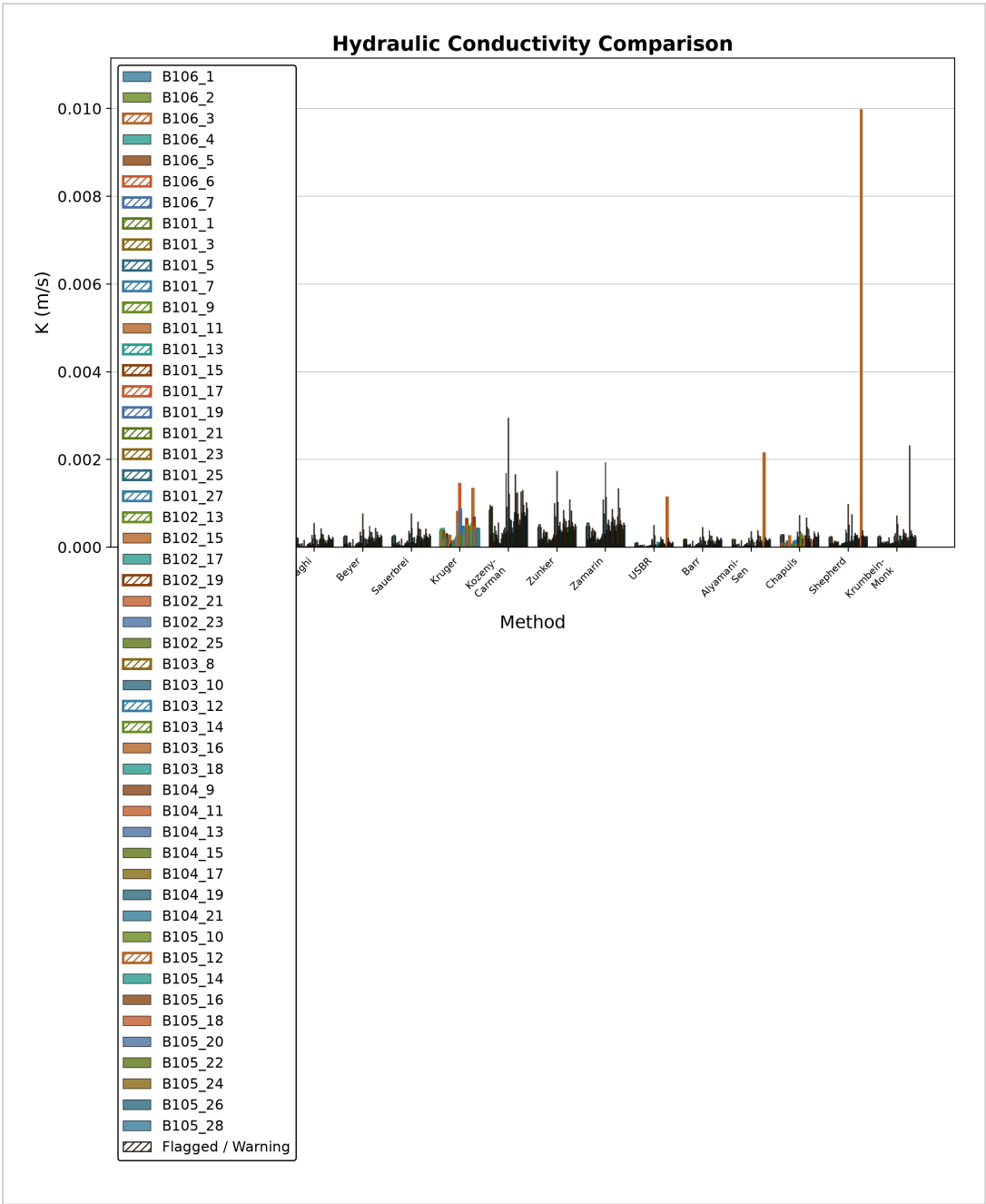
Analysis Components

Grain Size Comparison: Overlapping distribution curves allow visual assessment of particle size variations between samples.

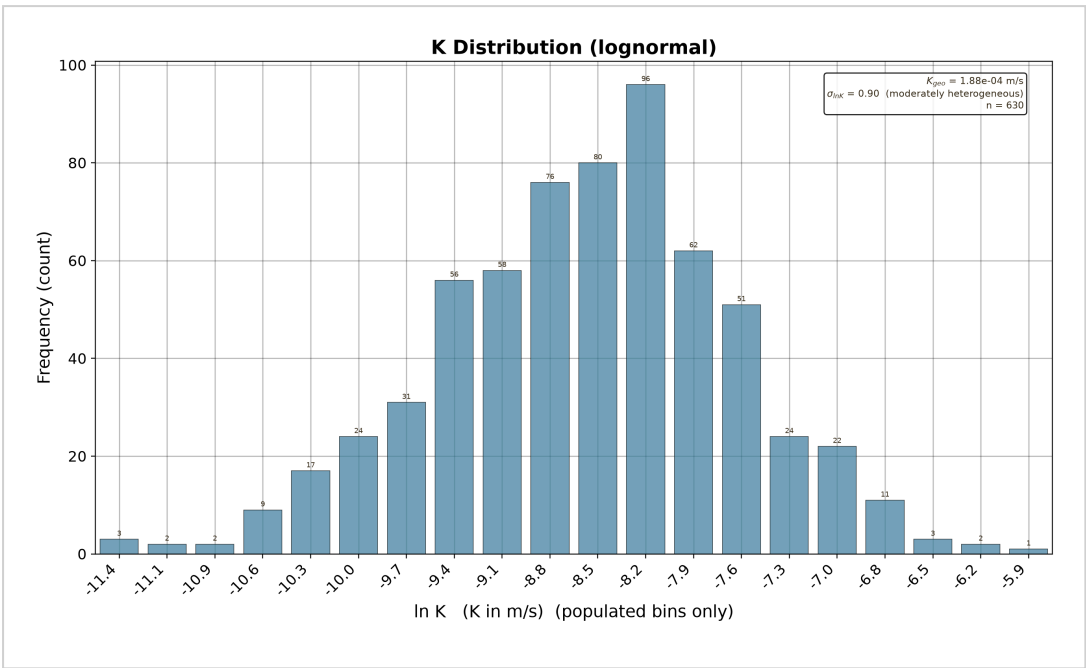
K-Value Comparison: Box plots and statistical summaries reveal the range and reliability of hydraulic conductivity estimates across samples.

Method Reliability: A reliability matrix shows which empirical methods are applicable for each sample, helping identify the most suitable estimation approaches.

Hydraulic Conductivity by Method



Hydraulic Conductivity Distribution (Lognormal)



Hydraulic Conductivity Distribution

